From Accounts to Alerts: EDA and Logistic Regression for Early Corporate Risk Prediction

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Outline

1. Motivation
2. Data
3. Methodology

Motivation

Fast-moving corporate world, risks can emerge quietly.

If we make decision from the news, it's too late.

- Early Warning Systems - not just useful, but essential.
  - Accessible – inputs are openly available
  - Trustworthy – the model can be explained clearly
  - Actionable – early alerts give time to respond
- Transparency and Interpretability
- Publicly Available Data

Graphical Abstracts

Data Preparation → Trainning & Test → Analysis & Improvement → Conclusion

Data Preparation

- EDA
- Imputation
- Winsorization
- Log-transform
- Data grouping & evaluation
- Missing value finding

Trainning & Test

- Label: Y = 1 if a SEC Form 25 becomes effective within 400 days after the financial statement date
- Features (X): 13 accounting variables
- Train: 2019–2023, Test: 2024

Analysis & Improvement

- Penalized logistic regression (L2 / Ridge) as baseline2
- Addresses class imbalance
- Calibration Analysis: evaluate AUC, PR-AUC, Brier, and calibration
- Post-hoc Improvements:
  - SMOTE for class imbalance
  - Platt scaling and Isotonic regression for probability calibration

Conclusion

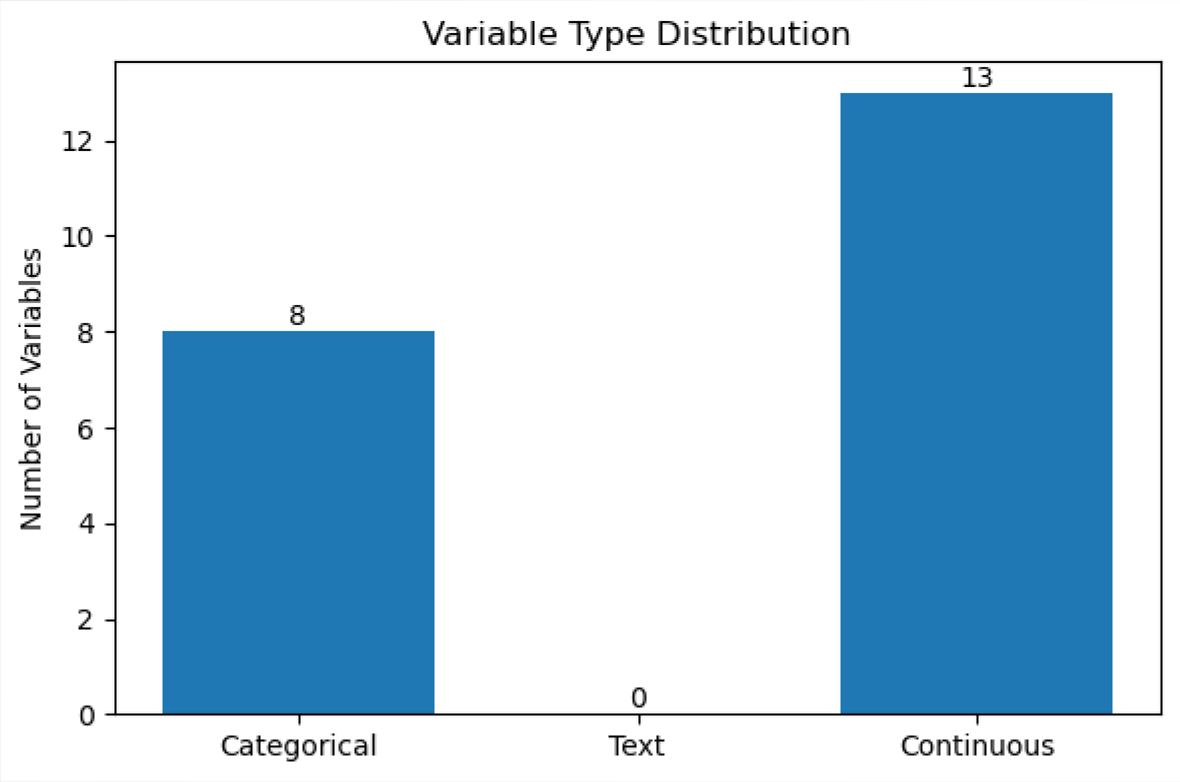
- Evaluation & Interpretation: discrimination, calibration, and coefficient analysis

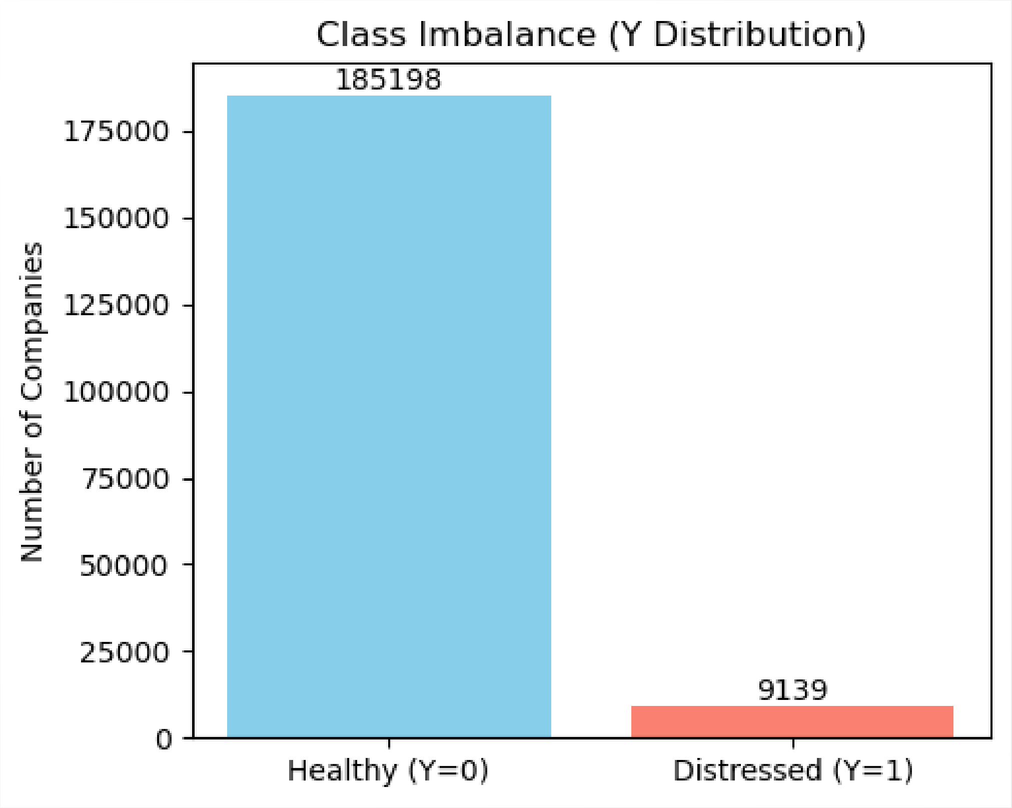
Overview

- USA annual financial statements (2019–2024) + SEC Form 25 delisting labels
- Label: Y=1 if a Form 25 becomes effective within 400 days after the financial statement date
- Train: 2019–2023, Test: 2024

Data Overview

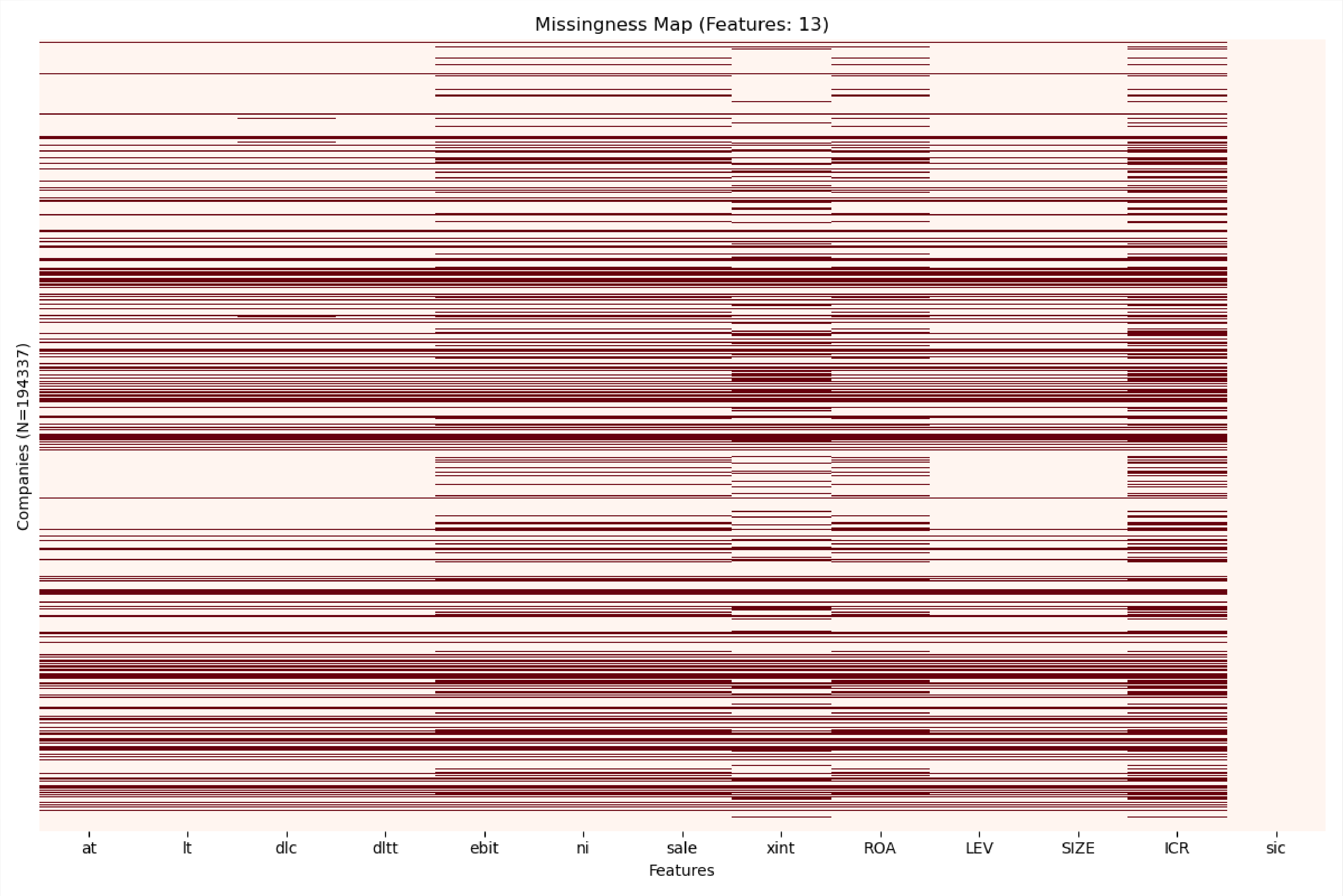
- **Source:** Fundamentals Annual (US Companies).
- **Scale:** 23,033 Companies / 194,337 Records.
- **Target (Y=1):** Form 25 filing within 400 days.
- **Imbalance:** Default rate is only ~4.7% (Highly Imbalanced).





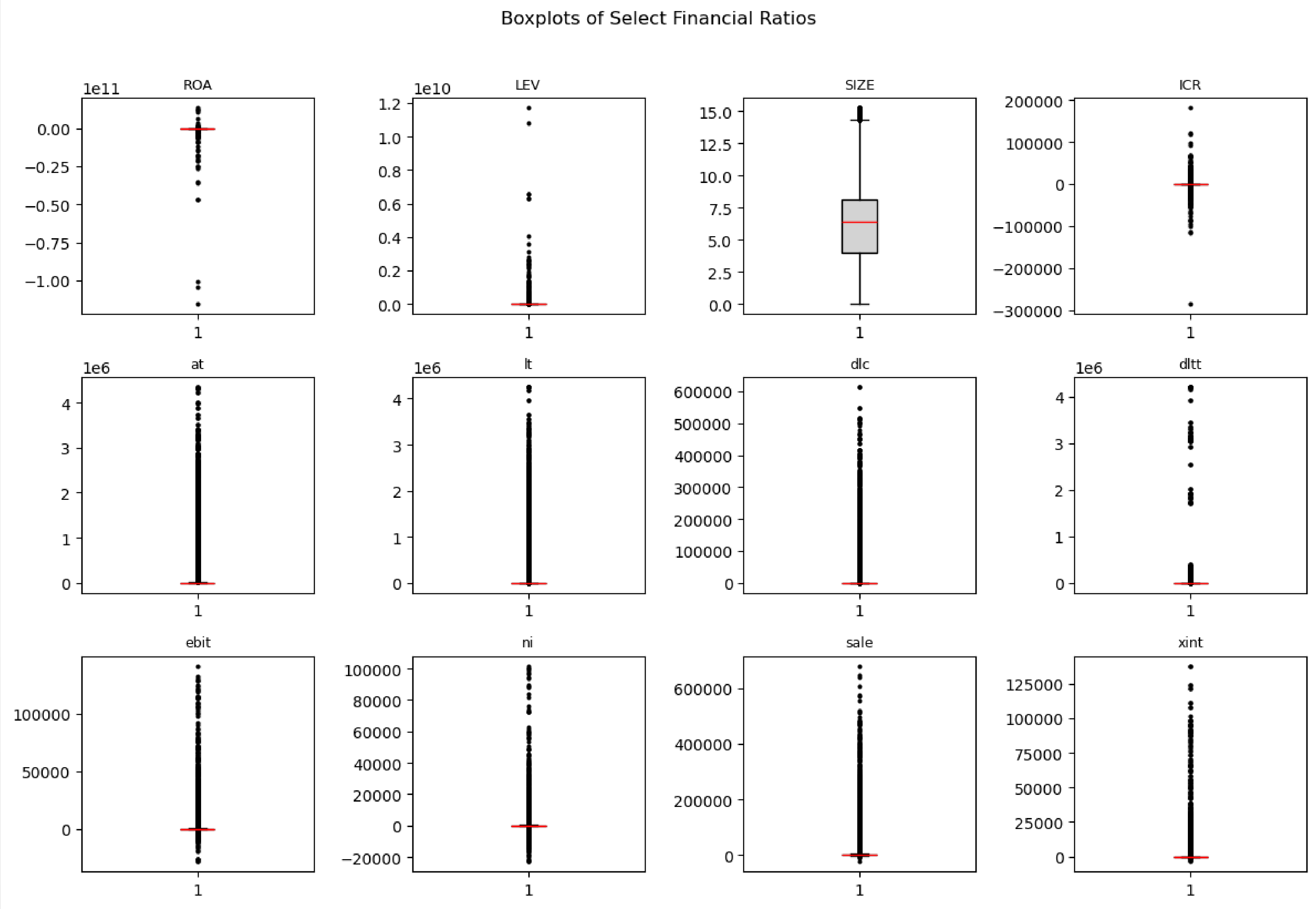
Data Quality & Missingness

- Average missing rate across all features: 18.17%
- Impute: SIC2×year median (train-only); else overall train median



Data

- heavy-tailed and asymmetric distributions



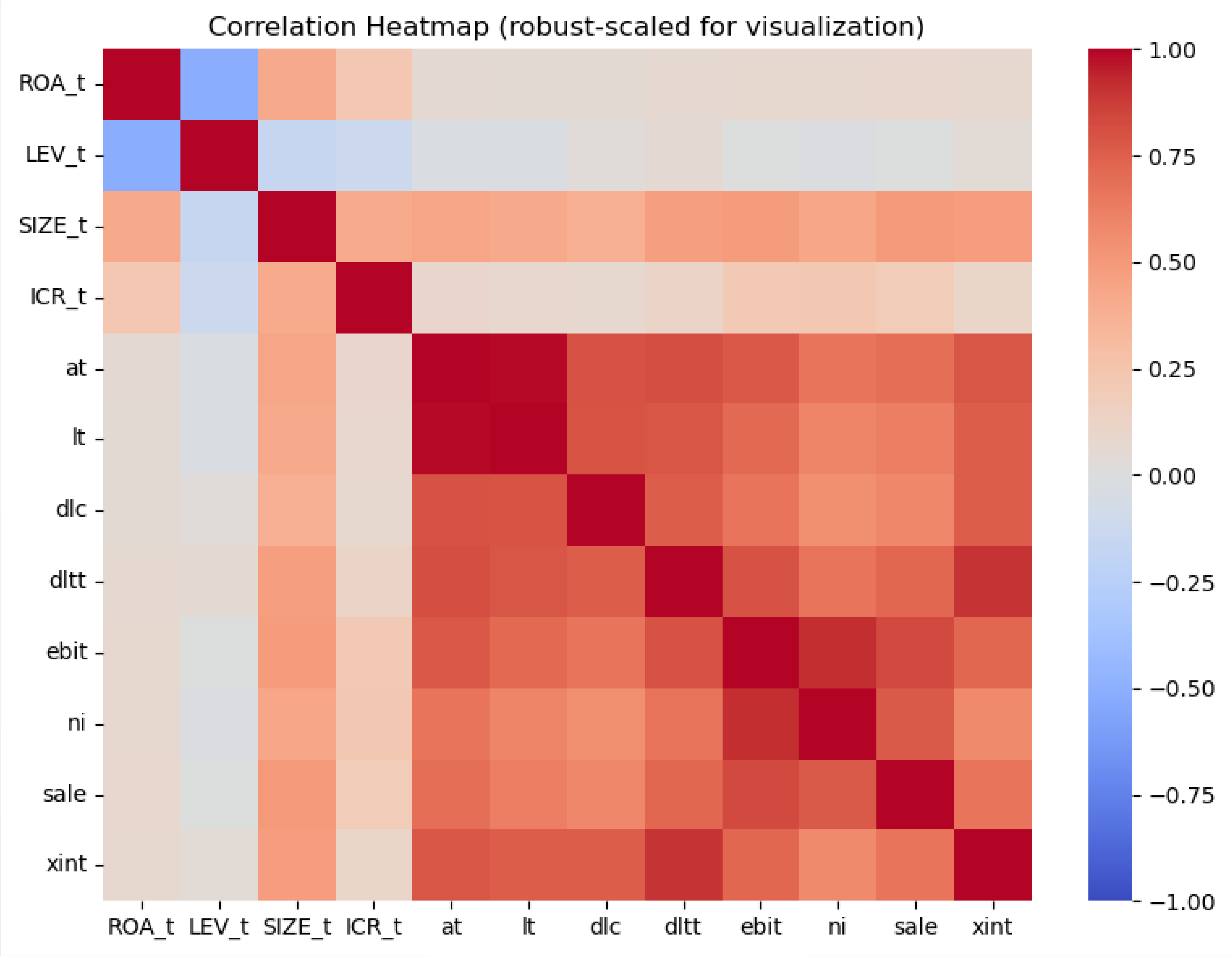
Data

- Most variables are highly skewed
- Several variables exhibit extreme kurtosis (>10)
- apply winsorization and log-transform before model estimation

	count	mean	std	min	25%	50%	75%	max	skewness	kurtosis
at	194337	15784.84	131619.6	0	143.705	782.563	3041.049	4349731	16.67897	335.2084
lt	194337	13364.29	123388	-0.103	50.374	421.99	1643.541	4255074	17.31502	360.7668
dlc	194337	1060.379	13402.19	-56	0.399	7.807	36.146	614237.4	22.55534	611.5345
dltt	194337	3011.604	54034.29	-0.023	2	58.802	975.806	4216909	57.62492	3679.058
ebit	194337	354.1788	2645.237	-28387	-0.389	3.9745	72.519	141290	22.55457	760.7693
ni	194337	180.8604	1596.6	-23119	-1.614	0.047	32.825	101832	25.60039	1139.399
sale	194337	2484.208	14554.11	-24954.7	44.032	166.73	430.529	680985	17.0715	430.196
xint	194337	146.4043	1910.341	-3775	1.092	6.6	49.351	137861	39.40192	1913.115
ROA	194337	-4476257	4.94E+08	-1.2E+11	-0.03658	0.000689	0.024949	1.35E+10	-174.64	35830.49
LEV	194337	846793.3	55324953	0	0.063014	0.188885	0.459505	1.17E+10	131.7885	22240.49
SIZE	194337	6.389006	2.611334	0	4.974621	6.663851	8.020287	15.28562	-0.36105	0.341554
ICR	194337	1.31E+09	1.68E+11	-2.1E+12	-0.44216	0.545308	2.908838	5.52E+13	237.4808	69517.85

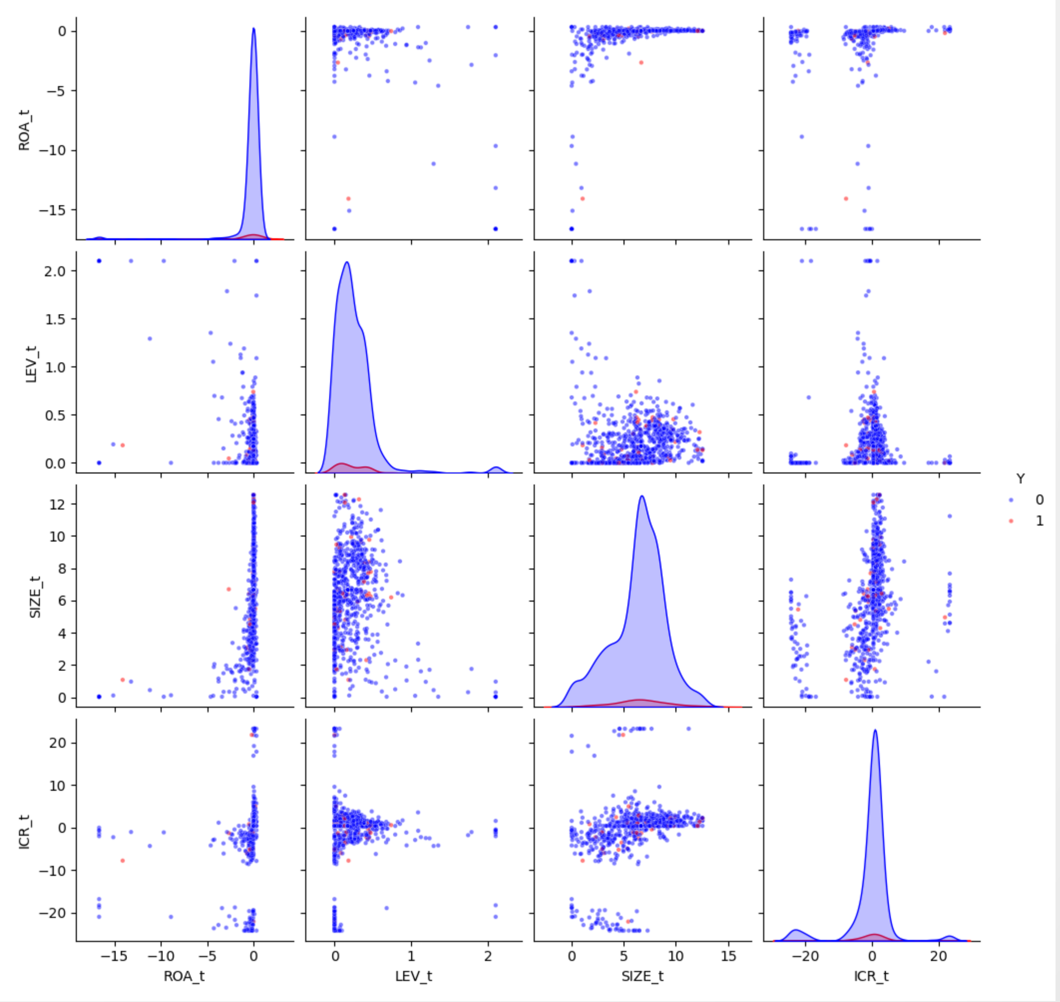
Data

- Positive correlations among size-related variables (*at*, *lt*, *sale*, *ni*)
- ROA inversely related to LEV (higher leverage, lower profitability)
- No major multicollinearity after data transformation



Feature Engineering

- Transformed 13 raw features into 4 predictors:
- Profitability: ROA (Return on Assets)
- Solvency: LEV (Leverage, Log-transformed)
- Coverage: ICR (Interest Coverage Ratio)
- Size: SIZE (Log of Total Assets)



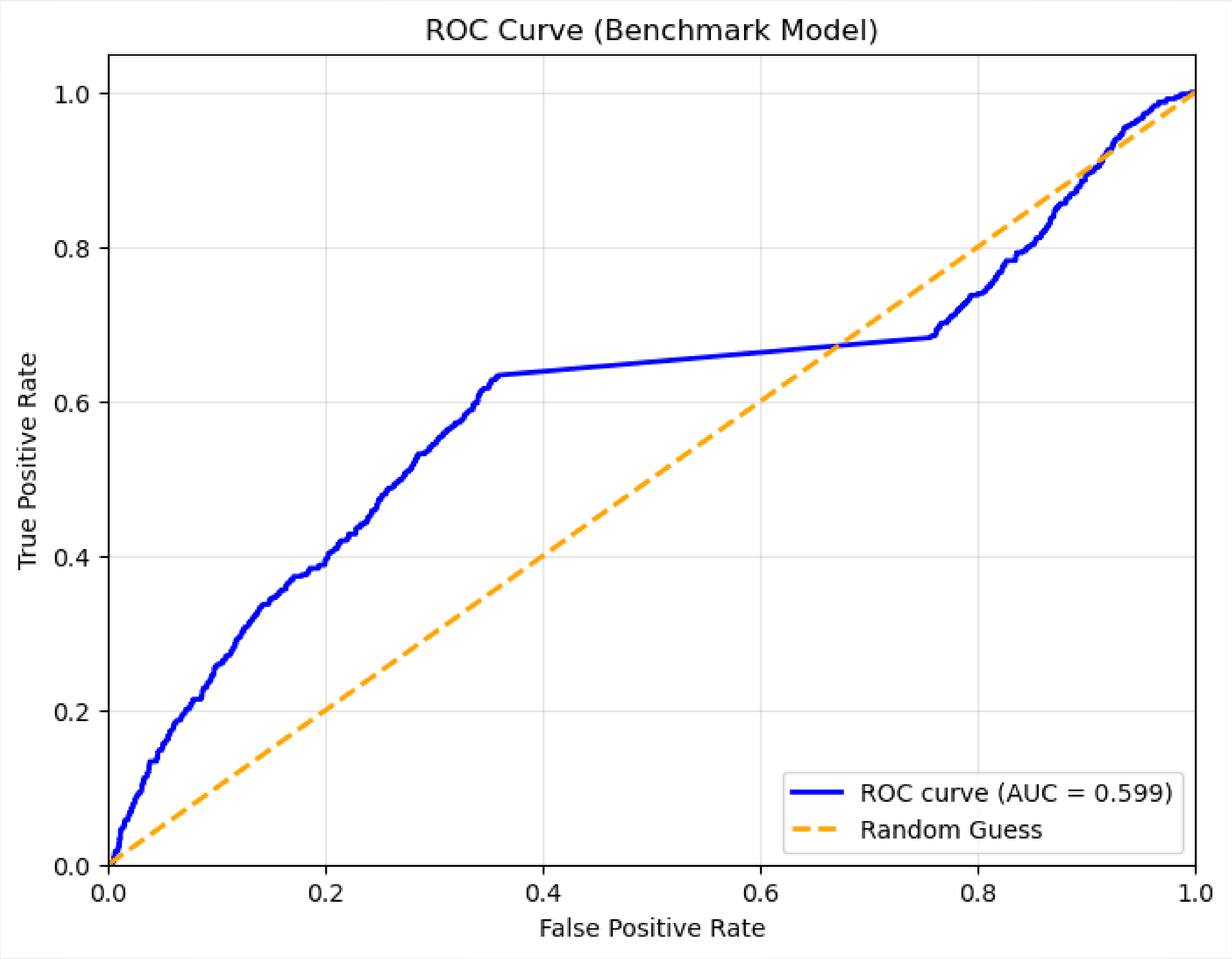
Methodology & Validation Strategy

- Benchmark Model: L2-Regularized Logistic Regression (Ridge)
- Training Split: 2019-2023 data.
- Testing Split: 2024 data (Out-of-Time Validation).
- Enhancements: SMOTE (for imbalance) + Isotonic Calibration (for probability accuracy).

Benchmark Model

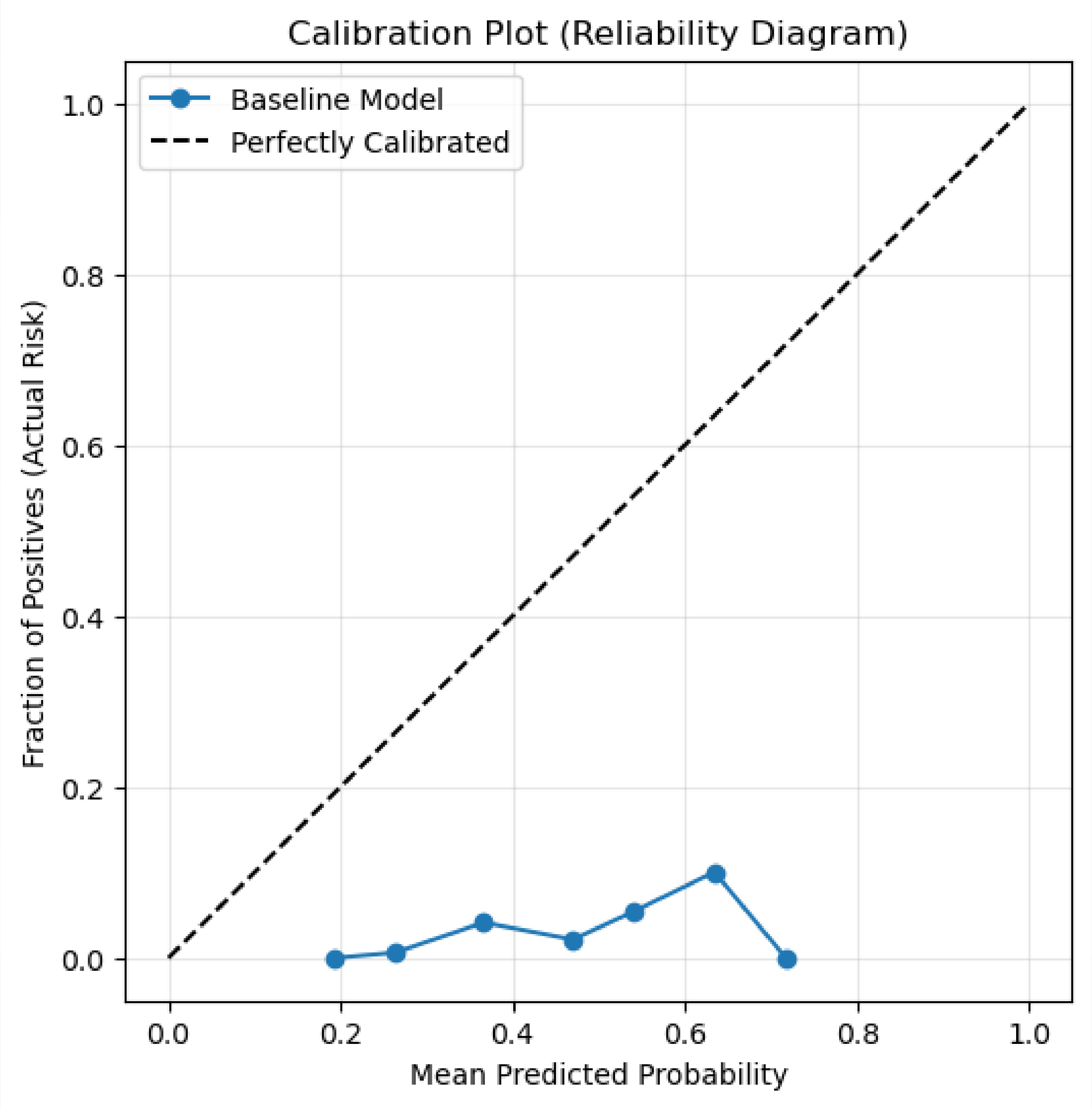
- Penalized logistic regression (L2 / Ridge) as baseline
- Mitigates overfitting and stabilizes coefficients
- Addresses class imbalance with class_weight="balanced“
- Evaluated by AUC, PR-AUC, Brier score, Calibration, and Youden’s J

Benchmark Model



Baseline L2-penalized logistic regression achieved AUC = 0.599, indicating moderate discriminatory power.

Benchmark Model (Calibration Analysis)



Predicted risk up to 0.6–0.7, but actual defaults stay <0.15
→ model overestimates risk and is poorly calibrated.

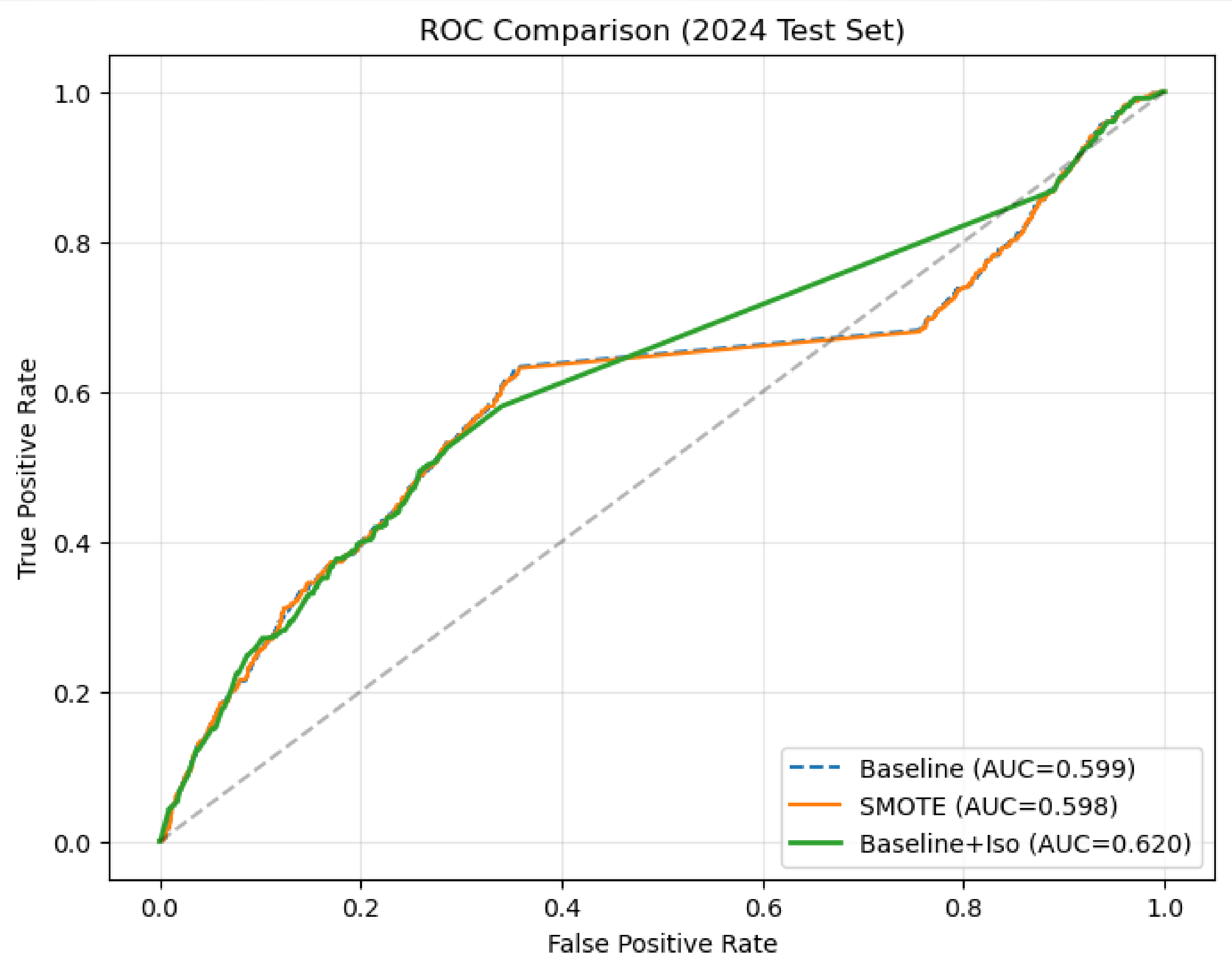
Design of Research

- Post-hoc Improvements:
  - *SMOTE* for class imbalance
  - *Platt scaling* and *Isotonic regression* for probability calibration
- Evaluation & Interpretation: discrimination (AUC, PR-AUC), calibration, and coefficient analysis

Post-hoc Improvements

- Post-hoc improvements include **SMOTE** for imbalance and **Platt/Isotonic** for probability calibration.

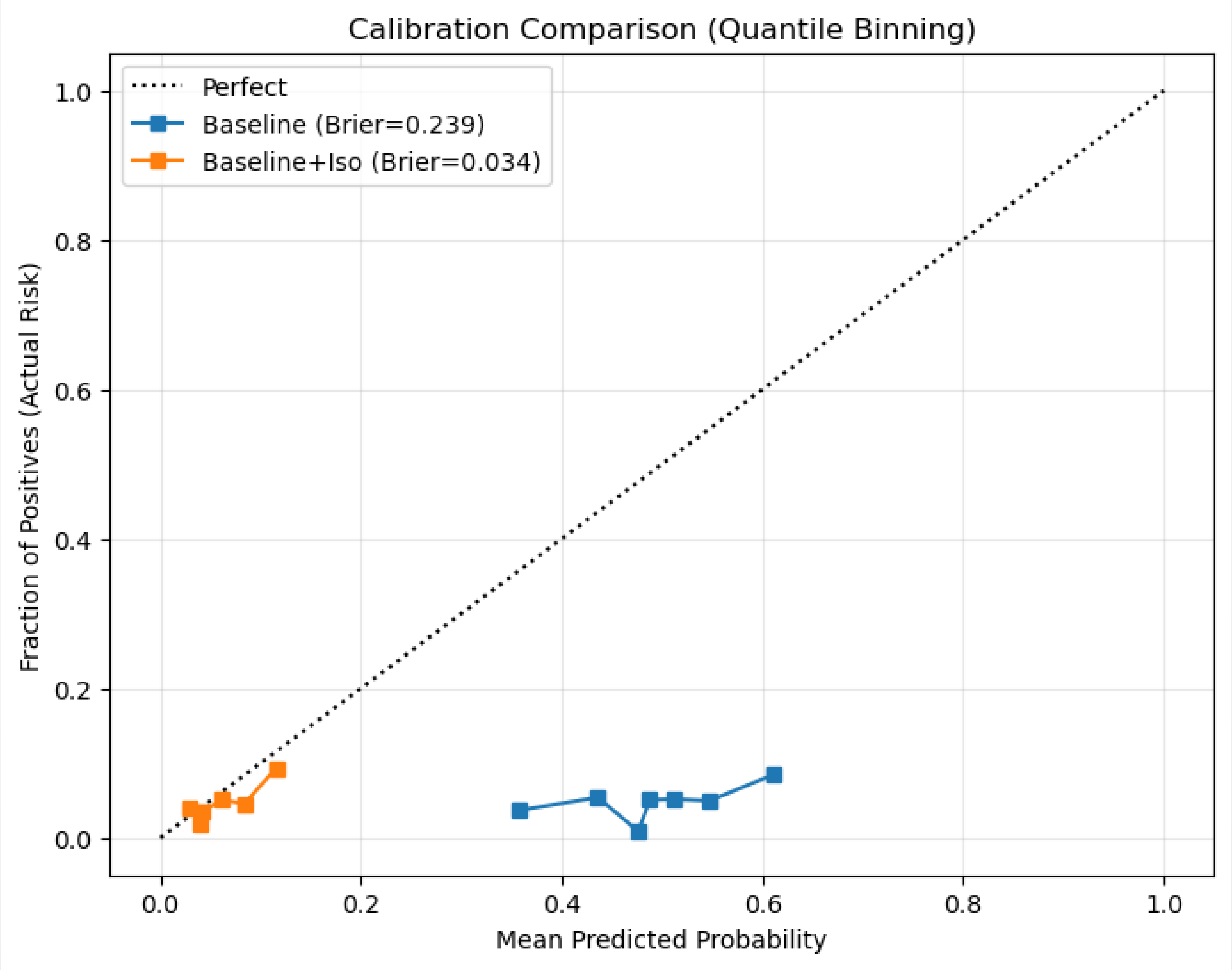
While SMOTE alone did not improve AUC, applying Isotonic Calibration increased the AUC to 0.620.



Post-hoc Improvements

- Post-hoc improvements include **SMOTE** for imbalance and **Platt/Isotonic** for probability calibration.

Post-hoc calibration (Isotonic) improves probability reliability, especially under imbalanced data (SMOTE).



Conclusion

- **Isotonic Calibration** significantly improved the reliability of risk probabilities
- This system can flag high-leverage firms (LEV) and liquidity issues (ICR) 400 days before delisting.

Feature	Coefficient
SIZE_t	0.316900
LEV_t	-0.047721
ROA_t	-0.119272
ICR_t	-0.205918

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